Lesson 11.1 - Hypothesis Testing - Difference of Two Means

> **Video:** Hypothesis Testing - Difference of Two Means - Student's t-Distribution & Normal Distribution by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

Setup

Two **independent** groups. Does their population mean differ?

$$H_0 : \mu_1 = \mu_2 \;\; (\mu_1 - \mu_2 = 0) \qquad H_a : \mu_1 \neq \mu_2 \;\; (\mu_1 - \mu_2 \neq 0)$$

Test statistic

$$t \text{ (or } z) = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under H_0, $\mu_1 - \mu_2 = 0$. The denominator is the **standard error of the difference**.

Sample sizes	Use
Small (< 30)	**t-distribution** (Welch), with the degrees of freedom below
Large (≥ 30)	**z** (normal distribution)

Degrees of freedom (Welch-Satterthwaite):

$$\text{df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \frac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

Example 1 - two teachers (t-test)

Do two teachers' classes score differently? $\alpha = 0.05$.

	Class 1	Class 2
n	15	12
$\bar{x}$	82	84
s	2.4	1.7

Hypotheses: $H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 \neq 0$ (two-tailed, $\alpha/2 = 0.025$).

Both samples are small → t-test.

Degrees of freedom:

- $s_1^2/n_1 = 5.76/15 = 0.384$ and $s_2^2/n_2 = 2.89/12 = 0.24083$
- Numerator: $(0.384 + 0.24083)^2 = (0.62483)^2 = 0.390417$

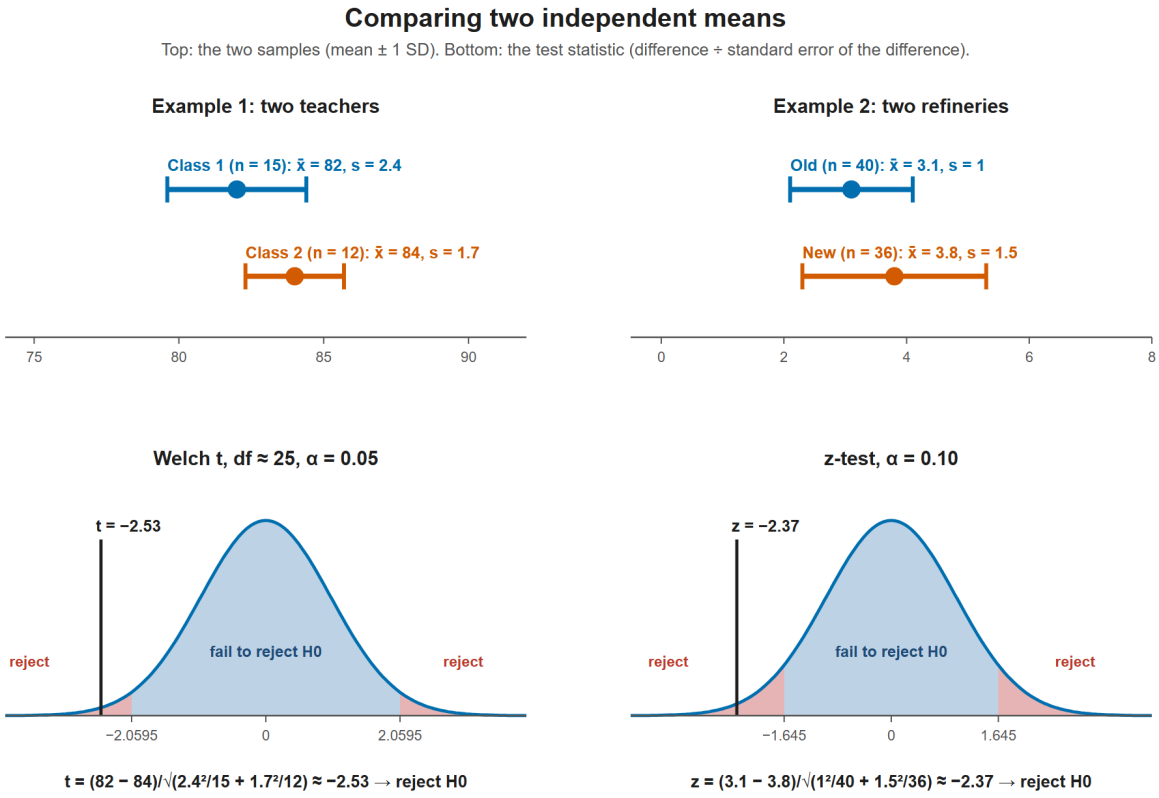
- Denominator: $\frac{0.384^2}{14} + \frac{0.24083^2}{11} = 0.010533 + 0.005273 = 0.015805$
- $\text{df} = 0.390417 / 0.015805 \approx 24.7$

The video rounds this to **25**, giving $t_{0.025,\,25} = 2.0595$. Many textbooks round df **down** to be conservative ($24 \to t = 2.064$). Both give the same decision here.

Test statistic:

$$t = \frac{82 - 84 - 0}{\sqrt{0.384 + 0.24083}} = \frac{-2}{0.79046} \approx -2.53$$

$-2.53 < -2.0595$, which is in the rejection region. **Reject H_0.** At the 5% level there is a significant difference between the two teachers' class means.



Top: the two samples in each example. Bottom: the difference in means divided by its standard error. In both examples the statistic lands in the left rejection region, so the means differ significantly.

Example 2 - two oil refineries (z-test)

Is there a difference in refining rate (liters per second)? $\alpha = 0.10$.

	Old factory	New factory
n	40	36
$\bar{x}$	3.1	3.8
s	1.0	1.5

Hypotheses: $H_0 : \mu_1 = \mu_2,\ H_a : \mu_1 \neq \mu_2$ (two-tailed).

Both samples are large → z-test. The confidence level is 90%, so the left area for the critical value is 0.95: $z = \pm 1.645$.

$$z = \frac{3.1 - 3.8}{\sqrt{\dfrac{1^2}{40} + \dfrac{1.5^2}{36}}} = \frac{-0.7}{\sqrt{0.025 + 0.0625}} = \frac{-0.7}{0.2958} \approx -2.37$$

$-2.37 < -1.645$ → **reject** H_0. At the 10% level there is a significant difference between the factories; the new one refines faster.

Key takeaways

- H_0: equal means (difference 0). Test statistic = (difference in sample means) ÷ $\sqrt{s_1^2/n_1 + s_2^2/n_2}$.
- Small samples → t with Welch's df. Large samples → z.
- Compare with the critical value(s), or with the p-value, as in the one-sample case.

Lesson 11.2 - How to Find Cohen's D to Determine the Effect Size

> **Video:** How to find Cohen's D to determine the Effect Size Using Pooled Standard Deviation by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

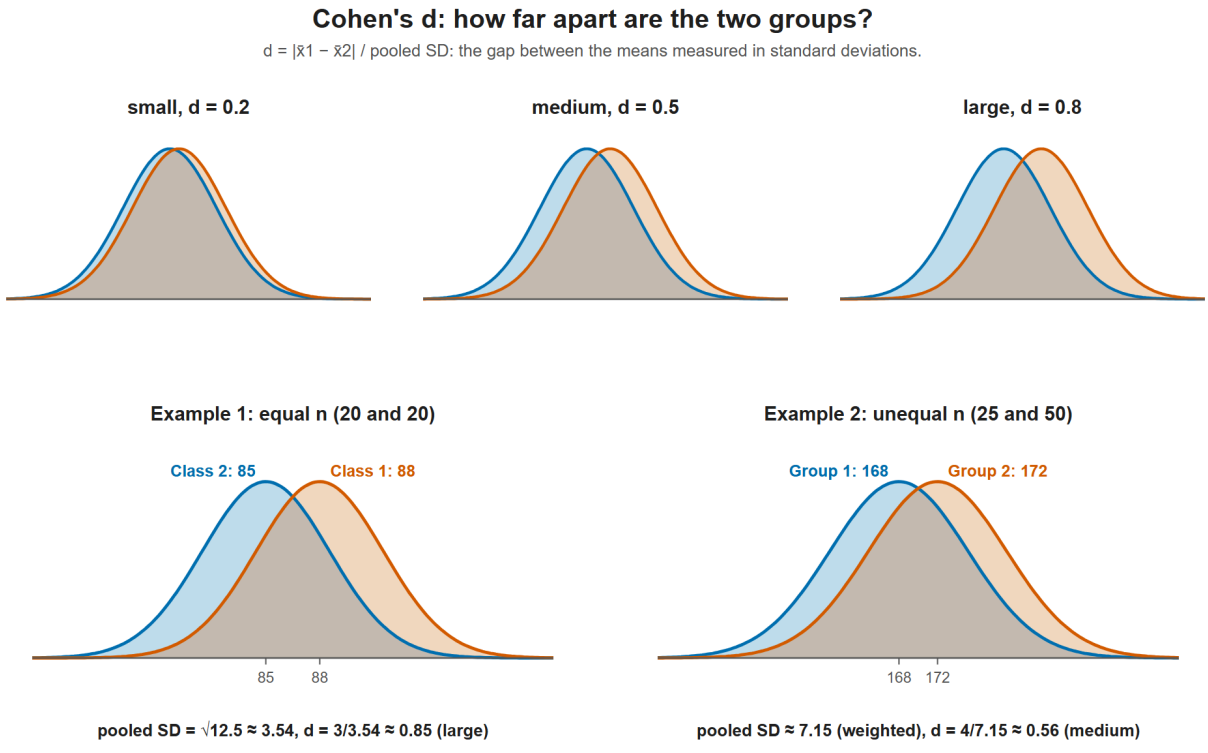
What Cohen's d measures

Cohen's d expresses the difference between two means **in standard-deviation units**:

$$d = \frac{|\bar{x}_1 - \bar{x}_2|}{s_{\text{pooled}}}$$

- $d = 1$: the means are 1 standard deviation apart.
- $d = 2$: 2 standard deviations apart.
- $d = 0.5$: half a standard deviation apart.

d	Effect size
≈ 0.2	Small
≈ 0.5	Medium
≥ 0.8	Large



Cohen's d measures how far apart the two curves sit, in units of the pooled standard deviation. The more the curves overlap, the smaller the effect. Example 1 (d ≈ 0.85) is a large effect; Example 2 (d ≈ 0.56) is medium.

Pooled standard deviation

General formula (use it when the sample sizes differ):

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Shortcut (only when $n_1 = n_2$, or nearly so):

$$s_{\text{pooled}} = \sqrt{\frac{s_1^2 + s_2^2}{2}}$$

Example 1 - two classes, equal sample sizes

	Class 1	Class 2
n	20	20
$\bar{x}$	88	85
s	3	4

Pooled SD (general formula):

$$s_p = \sqrt{\frac{19(3^2) + 19(4^2)}{20 + 20 - 2}} = \sqrt{\frac{171 + 304}{38}} = \sqrt{\frac{475}{38}} = \sqrt{12.5} \approx 3.5355$$

Shortcut (equal n): $\sqrt{\frac{9 + 16}{2}} = \sqrt{12.5} \approx 3.5355$, the same answer ✓

Cohen's d:

$$d = \frac{88 - 85}{3.5355} \approx 0.85 \;\Rightarrow\; \textbf{large effect}$$

Example 2 - two groups, unequal sample sizes

	Group 1	Group 2
n	25	50
$\bar{x}$ (lb)	168	172
s (lb)	5	8

Pooled SD (must use the general formula, because the sizes differ):

$$s_p = \sqrt{\frac{24(5^2) + 49(8^2)}{25 + 50 - 2}} = \sqrt{\frac{600 + 3136}{73}} = \sqrt{\frac{3736}{73}} \approx 7.154$$

The shortcut would give $\sqrt{(25 + 64)/2} \approx 6.671$, which is **wrong** here. Group 2 is twice as large, so its SD (8) should carry more weight, and the correct pooled SD (7.15) is closer to 8.

Cohen's d:

$$d = \frac{|168 - 172|}{7.154} = \frac{4}{7.154} \approx 0.56 \;\Rightarrow\; \textbf{medium effect}$$

Key takeaways

- d = |difference in means| ÷ pooled SD, a **unitless** measure of how big the difference is.
- Benchmarks: 0.2 small, 0.5 medium, 0.8+ large.
- Use the weighted pooled-SD formula unless the sample sizes are equal.
- A hypothesis test asks *whether* there is a difference. Cohen's d asks *how large* it is.

Lesson 11.3 - Hypothesis Testing with Two Proportions

> **Video:** Hypothesis Testing With Two Proportions by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

Formulas

Quantity	Formula
Sample proportions	$\hat{p}_1 = \frac{x_1}{n_1},\ \hat{p}_2 = \frac{x_2}{n_2}$
Pooled proportion	$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$
Test statistic	$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\,(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

Under $H_0 : p_1 = p_2$, the term $p_1 - p_2$ is **0**. The proportions are pooled because H_0 assumes both groups share one common proportion. Large samples → use the normal (z) distribution.

Example - defective laptops

Two batches of laptops were tested: 32 of 800 were defective in group 1 and 30 of 500 in group 2. Is the difference significant at $\alpha = 0.05$?

Hypotheses (the test is run because someone suspects a difference):

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

Sample proportions:

$$\hat{p}_1 = \frac{32}{800} = 0.04, \qquad \hat{p}_2 = \frac{30}{500} = 0.06$$

Critical values (two-tailed, $\alpha/2 = 0.025$ per tail, left area 0.975): $z = \pm 1.96$

Pooled proportion:

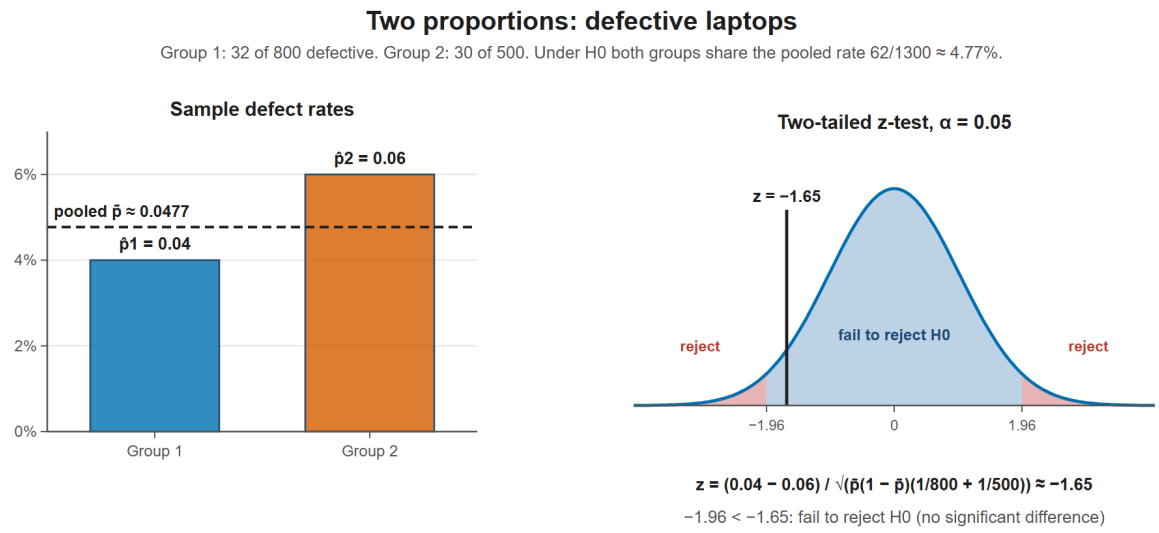
$$\bar{p} = \frac{32 + 30}{800 + 500} = \frac{62}{1300} \approx 0.04769$$

Test statistic:

$$z = \frac{0.04 - 0.06 - 0}{\sqrt{0.04769(1 - 0.04769)\left(\frac{1}{800} + \frac{1}{500}\right)}} = \frac{-0.02}{0.012149} \approx -1.65$$

$-1.96 < -1.65 < 0$, which is in the fail-to-reject region.

Fail to reject H_0. At the 5% level the difference in defect rates (4% vs 6%) is **not statistically significant**. This doesn't prove the rates are equal; the data just don't show a clear difference.



The two sample rates differ by 2 percentage points, but with these sample sizes that gives only z ≈ −1.65, which is inside ±1.96. The difference is not statistically significant.

Key takeaways

- $H_0 : p_1 = p_2$. Use the **pooled** proportion $\bar{p} = (x_1 + x_2)/(n_1 + n_2)$ in the standard error.
- $z = (\hat{p}_1 - \hat{p}_2)/\sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)}$.
- Compare with the critical value(s) or use a p-value. Fail to reject means "no significant difference found", not "proven equal".

Lesson 11.4 - Matched or Paired Samples

> **Video:** Matched or Paired Samples T-Test - Hypothesis Testing by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

When samples are paired

A **matched (paired) sample** measures the **same subjects twice**, for example before and after a treatment. Instead of comparing two separate groups, analyze the **difference** for each subject:

$$x_d = \text{after} - \text{before}$$

Formulas

Quantity	Formula
Mean difference	$\bar{x}_d = \dfrac{\sum x_d}{n}$
SD of the differences	$s_d = \sqrt{\dfrac{\sum (x_d - \bar{x}_d)^2}{n-1}}$ (or `=STDEV.S(...)` in Excel)
Test statistic	$t = \dfrac{\bar{x}_d - \mu_d}{s_d/\sqrt{n}}$, $\text{df} = n-1$
Confidence interval	$\bar{x}_d \pm t\,\dfrac{s_d}{\sqrt{n}}$

Example - a weight-loss program

Is the program effective at reducing weight? $\alpha = 0.05$, 10 subjects.

Subject	Before (lb)	After (lb)	x_d = after − before
1	185	169	−16
2	192	187	−5
3	206	193	−13
4	177	176	−1
5	225	194	−31
6	168	171	+3
7	256	228	−28
8	239	217	−22
9	199	204	+5
10	218	195	−23

A negative difference means the subject lost weight.

Hypotheses. An effective program means the mean difference is **negative**:

$$H_0 : \mu_d \geq 0 \qquad H_a : \mu_d < 0 \quad \text{(left-tailed)}$$

Sample statistics:

$$\bar{x}_d = \frac{-131}{10} = -13.1, \qquad s_d = \sqrt{\frac{1526.9}{9}} \approx 13.025$$

Critical value: left-tailed, df $= 9$, $\alpha = 0.05$ gives $t = -1.833$.

Test statistic:

$$t = \frac{-13.1 - 0}{13.025/\sqrt{10}} = \frac{-13.1}{4.119} \approx -3.18$$

$-3.18 < -1.833$, which is in the rejection region. **Reject H_0.** There is sufficient evidence at the 5% level that the program reduces weight.

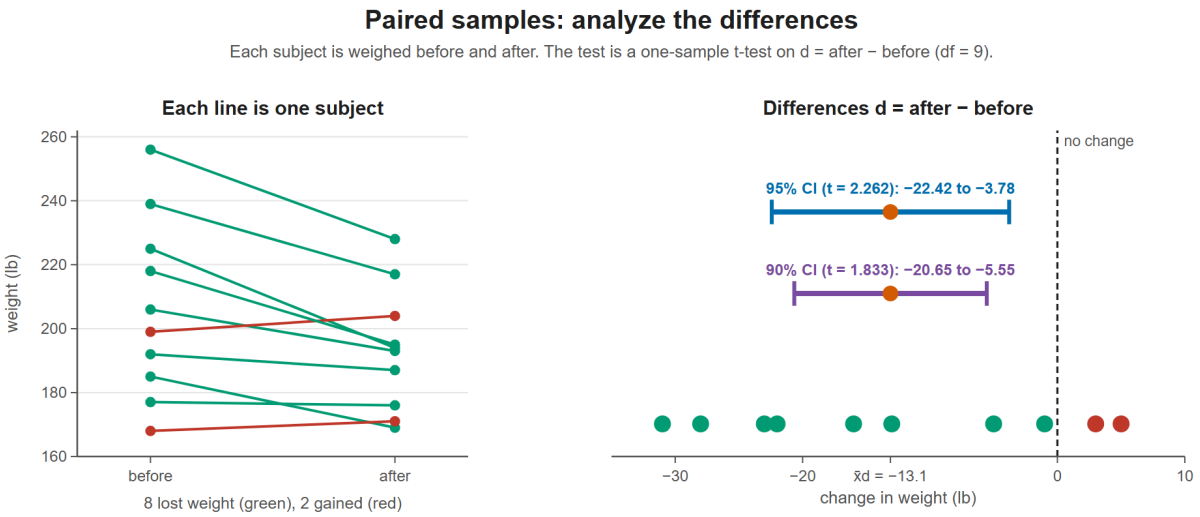
Confidence interval for μ_d

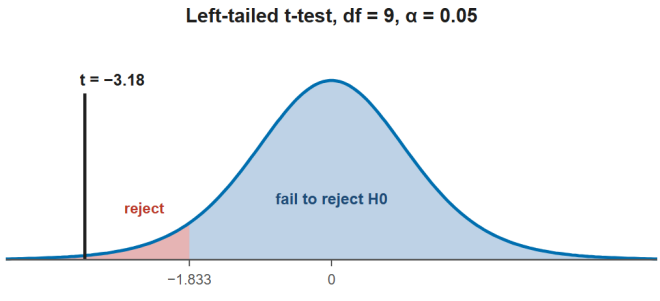
Using $t = 1.833$ (as in the video):

$$-13.1 \pm 1.833\,(4.119) = -13.1 \pm 7.55 \;\Rightarrow\; (-20.65,\ -5.55)$$

> **Correction:** 1.833 is the critical value with **5% in one tail**, so a two-sided interval built from it is a **90%** confidence interval, not 95%. For a **95%** interval, use $t_{0.025,\,9} = 2.262$: $-13.1 \pm 2.262(4.119) = -13.1 \pm 9.32 \Rightarrow (-22.42,\ -3.78)$.

Either way, the whole interval is **below 0**: there are no zero or positive values, which supports rejecting H_0. The margin of error is 7.55 for the 90% interval, or 9.32 for 95%.





t = −13.1 / (13.025/√10) ≈ −3.18 < −1.833 → reject H0: the program reduces weight

Top left: most subjects lost weight. Top right: the ten differences, with the 90% and 95% intervals for the mean difference; both sit entirely below 0. Bottom: t = −3.18 is beyond the critical value −1.833, so H0 is rejected.

Key takeaways

- Paired data → work with the **differences** x_d and run a one-sample t-test on them.
- df $= n - 1$, where n is the number of **pairs**.
- Choose the direction of H_a to match the claim (e.g. "reduces" → $\mu_d < 0$ when x_d = after − before).
- Match the CI's t to its confidence level: two-sided 95% uses $\alpha/2 = 0.025$.